(Universidade de São Paulo)
Pêndulo Simples e Pêndulo Duplo

Métodos Numéricos em Equações Diferenciais

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1 Pêndulo Simples

1.1 EDO

$$\mathbf{u}(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix} = \begin{pmatrix} q(t) \\ q'(t) \end{pmatrix} = \begin{pmatrix} q(t) \\ p(t) \end{pmatrix}, \quad q''(t) + \sin(q(t)) = 0 \iff p'(t) = -\sin(q(t))$$

$$\mathbf{u}'(t) = \begin{pmatrix} q'(t) \\ p'(t) \end{pmatrix} = \begin{pmatrix} p(t) \\ -\sin(q(t)) \end{pmatrix} = \begin{pmatrix} u_2(t) \\ -\sin(u_1(t)) \end{pmatrix} = \begin{pmatrix} f_1(\mathbf{u}(t), t) \\ f_2(\mathbf{u}(t), t) \end{pmatrix} = \mathbf{f}(\mathbf{u}(t), t)$$

$$\mathbf{u}^0 = \mathbf{u}(0) = \begin{pmatrix} q(0) \\ p(0) \end{pmatrix} = \begin{pmatrix} \frac{\pi}{4} \\ 0 \end{pmatrix}$$

1.2 Métodos

Para $h > 0$, $t_n = hn$, $\mathbf{U}^n = \begin{pmatrix} U_1^n \\ U_2^n \end{pmatrix} \approx \mathbf{u}(t_n)$ e $\mathbf{f}^n = \mathbf{f}(\mathbf{U}^n, t_n)$ temos:

1. Euler explícito: $\mathbf{U}^{n+1} = \mathbf{U}^n + h\mathbf{f}^n$;
2. RK4: $\begin{cases} K_1 = \mathbf{f}(\mathbf{U}^n, t_n) \\ K_2 = \mathbf{f}(\mathbf{U}^n + \frac{h}{2}K_1, t_n + \frac{h}{2}) \\ K_3 = \mathbf{f}(\mathbf{U}^n + \frac{h}{2}K_2, t_n + \frac{h}{2}) \\ K_4 = \mathbf{f}(\mathbf{U}^n + hK_3, t_n + h) \\ \mathbf{U}^{n+1} = \mathbf{U}^n + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4) \end{cases}$;
3. Euler implícito: $\mathbf{U}^{n+1} = \mathbf{U}^n + h\mathbf{f}^{n+1}$

1.2.1 Euler implícito

$$\mathbf{U}^{n+1} = \mathbf{U}^n + h\mathbf{f}^{n+1} \iff \mathbf{F}(\mathbf{U}) = \mathbf{U} - \mathbf{U}^n - h\mathbf{f}(\mathbf{U}, t_n) = 0$$

$$\mathbf{F}'(\mathbf{U}) = \frac{d\mathbf{F}}{d\mathbf{U}}(\mathbf{U}) = I_n - h\frac{d\mathbf{f}}{d\mathbf{U}}(\mathbf{U}, t_n)$$

onde I_n é a matriz identidade $n \times n$ e

$$\frac{d\mathbf{f}}{d\mathbf{U}}(\mathbf{U}, t_n) = \left.\begin{pmatrix} \frac{\partial f_1}{\partial U_1} & \frac{\partial f_1}{\partial U_2} \\ \frac{\partial f_2}{\partial U_1} & \frac{\partial f_2}{\partial U_2} \end{pmatrix}\right|_{(\mathbf{U}, t_n)} = \begin{pmatrix} 0 & 1 \\ -\cos(U_1) & 0 \end{pmatrix}$$

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$$\mathbf{F}'(\mathbf{U}) = \begin{pmatrix} 1 & -h \\ h\cos(U_1) & 1 \end{pmatrix}$$

Assim, com $\mathbf{U} = \mathbf{U}^{n+1}$ e um chute inicial $\mathbf{U}^{(0)}$, temos o método de Newton, para $k = 0, 1, \ldots$ e com critério de parada $\|\boldsymbol{\delta}_k\|_2 < 10^{-7}$:

$$\begin{cases} \mathbf{F}'(\mathbf{U}^{(k)})\boldsymbol{\delta}_k = -\mathbf{F}(\mathbf{U}^{(k)}) \\ \mathbf{U}^{(k+1)} = \mathbf{U}^{(k)} + \boldsymbol{\delta}_k \end{cases}$$